

Teacher:	Ray Joshua Balingkit	**Subject:**	Computer Science
Week:	6 (Term 1)	**Year:**	10-11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will learn to use decomposition and abstraction to understand a problem. They will also learn to apply decomposition and abstraction to plan a solution. Teacher demonstrations, guided examples and practical tasks will help students explain and apply these ideas.

End-of-Week Assessment/Check: How will you know they got it?

Teacher questioning, class discussion, the completed textbook activities and the final checks in both lessons. Explain one subproblem and one detail removed through abstraction. Explain how decomposition and abstraction improved the solution. Review completed work and use the final checks to identify misconceptions.

LESSON SLOTS - MONDAY & FRIDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	28 September 2026	By the end of this lesson, students will be able to use decomposition and abstraction to understand a problem.	**H:** Recall prior learning with a short question or prediction linked to the lesson topic (5 minutes). **ID:** Break a school booking problem into smaller tasks and identify essential details. Use textbook pages 23-31 and explain the relevant vocabulary (10 minutes). **WD:** Complete a second example together. Ask students to explain each step and check for misconceptions (10 minutes). **YD:** Decompose a new problem and separate relevant from unnecessary information. Record the solution and check it against the task requirements (15 minutes). **C:** Explain one subproblem and one detail removed through abstraction (5 minutes).	**Support:** Provide a vocabulary guide, a worked example and a partially completed task. Use prompts and paired rehearsal before independent work. **Extension:** Apply the concept to a new example, justify the choices and explain a limitation or alternative solution.



2	02 October 2026	By the end of this lesson, students will be able to apply decomposition and abstraction to plan a solution.	**H:** Recall prior learning with a short question or prediction linked to the lesson topic (5 minutes). **ID:** Connect subproblems to an algorithm and model a Unit 1 question. Use textbook pages 23-31 and explain the relevant vocabulary (10 minutes). **WD:** Complete a second example together. Ask students to explain each step and check for misconceptions (10 minutes). **YD:** Plan a solution and complete selected Unit 1 questions in book order. Record the solution and check it against the task requirements (15 minutes). **C:** Explain how decomposition and abstraction improved the solution (5 minutes).	**Support:** Provide a vocabulary guide, a worked example and a partially completed task. Use prompts and paired rehearsal before independent work. **Extension:** Apply the concept to a new example, justify the choices and explain a limitation or alternative solution.
3				Support: Extension:
4				Support: Extension:



MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel International GCSE Computer Science Student Book, printed pages 23-31
- Teacher computer and classroom display
- Student notebooks, algorithm worksheets and drawing materials
- Relevant textbook examples, activities and questions



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?